

Aankit Das

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On F1 OPT STEM, Eligible to work in the US without sponsorship | TX

Work History

AI Engineer at Bantrly, Dallas, TX

Mar 2026 to Present

- Built the speech and language systems behind a K-12 learning app by turning student audio into text and scoring it against teaching rubrics
- Evaluated neural text-to-speech engines pytttsx3, Kokoro, edge-tts, Chatterbox, and Chirp against a rubric of intelligibility, naturalness, pacing, and latency per K-12 grade band, producing cost-quality comparisons that guided engine selection
- Designed an LLM-as-judge evaluation pipeline combining structured JSON output, retry logic, and cosine-similarity consistency scoring via nomic-embed-text embeddings across 468 controlled experiments on 3 open-source models, quantifying quality, consistency, and hallucination trade-offs

Machine Learning Researcher at University of Texas at Dallas, Richardson, TX

Jun 2025 to Mar 2026

- Built and optimized neural network pipelines to extract clean signal from noisy space telescope data for an NSF-funded cosmology research project
- Contributed to the CMB-ML open-source research organization, supporting ongoing cosmology and machine learning research toward a planned ICCV or CVPR submission
- Processed and managed 1,600 high-resolution CMB simulated maps totaling 24GB across multiple splits for model training
- Ran ablation studies across dropout, batch normalization, loss function variants, and activation functions, comparing MSE, normalized MSE, and Wishart losses alongside ReLU and GELU activations

Data Scientist at SEI Investments Company, Kolkata, WB

Oct 2021 to Aug 2023

- Built automation tools and data workflows for a financial services team, turning manual document and reconciliation processes into faster digital systems
- Developed a Python-based prototype for automating document-filling using NLP libraries and OCR technology, reducing manual processing by 50 percent and improving data availability
- Architected a reconciliation workflow for diverse investor profiles using Alteryx, boosting efficiency and integration across cross-functional teams
- Implemented error-handling mechanisms and documented the algorithm design, implementation steps, and performance metrics, improving reliability and knowledge sharing across the team
- Built Tableau dashboards for large-scale datasets, giving business stakeholders actionable insights for decision-making

Research Assistant at University of Nebraska-Lincoln, Remote

May 2021 to Jun 2023

- Developed a machine learning segmentation model and benchmark dataset for tracking early plant growth from time-lapse images

- Collaborated with international research teams to develop EmergeNet, a segmentation model for agricultural data that achieved 99.4 percent IoU against a 69 percent baseline for a standard UNet
- Developed the UNL-MED benchmark dataset, curating top-view time-lapse images of maize coleoptiles from pre-emergence through roughly one inch of growth to evaluate EmergeNet
- Applied optimization techniques to monitor maize coleoptile growth, reaching a 0.99 correlation between predicted and observed metrics and publishing the findings in Frontiers and an e-book

Research Assistant at Jadavpur University, Kolkata, WB

Apr 2020 to Aug 2021

- Researched nature-inspired optimization algorithms for audio-based language identification, publishing the results in IEEE journals
- Innovated nature-inspired meta-heuristic feature selection algorithms, BBA-LAHC and HS-NMR, for spoken language identification, achieving over 98 percent test accuracy
- Published two IEEE journal papers documenting advances in NLP and audio signal processing research

Projects

Document Intelligence Platform, Production RAG System

- Built a retrieval-augmented generation system that lets users ask questions in plain language and get answers grounded in their own documents
- Built a semantic chunking pipeline with 500-token chunks and 50-token overlap, using PyPDF2 extraction to generate 768-dimensional embeddings for cosine similarity retrieval under 10ms latency on over 100 document chunks
- Demonstrated a cost-optimization strategy, pairing local embeddings for privacy and efficiency with free-tier LLM APIs for fast inference and persistent vector storage for retrieval performance

Education & Certificates

- MS, Computer Engineering, University of Texas at Dallas (GPA - 3.8/4.0) May 2025
- BTech, Electronics and Communication Engineering, University of Calcutta (GPA - 3.7/4.0) Sept 2020

Publications

- **Das, A.** A hybrid meta-heuristic feature selection method for identification of Indian spoken languages from audio signals. IEEE Access.
- **Das, A.** EmergeNet: A novel deep-learning based ensemble segmentation model for emergence timing detection of coleoptile. Frontiers in Plant Science.
- **Das, A.** Hybrid feature selection method based on harmony search and naked mole-rat algorithms for spoken language identification from audio signals. IEEE Access.
- **Das, A.** Flowerphenonet: Automated flower detection from multi-view image sequences using deep neural networks for temporal plant phenotyping analysis. MDPI.